

VISUALIZING DATA DISTRIBUTIONS AND ANALYZING RELATIONSHIPS

*Statistical Analysis of Video Game Player Metrics and Health Monitoring
Indicators*

Student Information

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2. Project Overview

Objective of the Project: This capstone project conducts a rigorous statistical evaluation of player engagement and satisfaction metrics in video games, alongside physiological and behavioral public health monitoring indicators. Using exploratory data analysis (EDA), box plot visualization, and advanced correlation modeling, this project evaluates the distributional characteristics, variability, central tendencies, and outlier behaviors of real-world variables. The underlying analysis highlights how selecting appropriate statistical methods specifically matching correlation coefficients to variable measurement scales is essential for robust, valid, and actionable data-driven scientific conclusions.

Datasets Selected: To achieve a robust comparative perspective, both available datasets were fully selected and analyzed for this project:

1. Video Game Reviews Dataset: Consists of 60 individual player profiles analyzing Hours Played and subjective Review Scores.
2. Health Monitoring Dataset: Consists of 120 validated physiological profiles measuring Age, Smoking Status, and Systolic Blood Pressure.

Tools Used: Microsoft Excel was used for data organization, statistical calculations, correlation analysis, skewness calculations, and box plot visualizations. Microsoft Word was used for report preparation.

3. Exploratory Data Analysis (EDA)

3.1 Dataset Description

A foundational review of the datasets highlights distinct structural characteristics and variable types:

- Video Game Reviews Dataset (N = 60): Investigates the relationship between playtime and satisfaction. It features three attributes: Player ID (a nominal identifier/categorical variable), Hours Played (a continuous numerical variable, representing the cumulative gameplay duration in hours), and Review Score (a continuous numerical variable, representing the user-assigned game score on a scale of 0 to 100).
- Health Monitoring Dataset (N = 120): Evaluates cardiovascular risk factors across distinct age brackets. It features four attributes: Participant ID (a nominal identifier/categorical variable), Age (a continuous numerical variable in years), Blood Pressure (mmHg) (a continuous numerical variable representing systolic blood pressure), and Smoking Status (a binary categorical/dichotomous variable with values 'Yes' or 'No'). Age Group is a derived ordinal categorical variable categorizing participants into 20s, 30s, 40s, 50s, and 60+ groups.

3.2 Summary Statistics

Descriptive statistics were computed for all relevant numerical attributes across both datasets. Tables 1 and 2 present these summaries, capturing central tendency, dispersion, and range parameters.

Table 1: Descriptive Statistics for Video Game Reviews Dataset (N = 60)

Numerical Variable	Mean	Median	Mode	Minimum	Maximum	Std. Dev.
Review Score (out of 100)	80.62	80.50	89.00	63.30	100.00	9.91
Hours Played (playtime)	16.33	16.00	16.00	5.00	45.00	7.71

Table 2: Descriptive Statistics for Health Monitoring Dataset (N = 120)

Numerical Variable	Mean	Median	Mode	Minimum	Maximum	Std. Dev.
Blood Pressure (mmHg)	128.71	128.00	112, 126, 128, 141	98.00	166.00	14.94
Participant Age (years)	47.28	48.00	Multimodal	20.00	77.00	16.73

3.3 Five-Number Summary

The five-number summary provides a robust framework to understand the position, spread, and outlier presence within continuous distributions. Table 3 presents the five-number summaries (comprising Minimum, first quartile [Q1], Median [Q2], third quartile [Q3], and Maximum), along with the Interquartile Range (IQR = Q3 – Q1) and outlier counts, for player review scores and systolic blood pressure categorized by age group.

Table 3: Comprehensive Five-Number Summaries and Outlier Metrics Across Datasets

Target Group / Metric	Minimum	Q1 (25%)	Median (Q2)	Q3 (75%)	Maximum	IQR	Outliers
Review Score (N=60)	63.30	71.75	80.50	89.00	100.00	17.25	0.00
BP - 20s Group (N=24)	98.00	115.38	126.50	138.00	151.00	22.63	0.00
BP - 30s Group (N=20)	108.00	126.75	131.20	135.40	166.00	8.65	4.00
BP - 40s Group (N=18)	104.00	120.00	138.50	143.25	163.00	23.25	0.00
BP - 50s Group (N=25)	100.30	117.00	126.00	136.00	148.60	19.00	0.00

BP - 60+ Group (N=33)	105.60	113.00	126.00	141.00	161.00	28.00	0.00
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4. Box Plot Visualizations

4.1 Single Box Plot Analysis: Player Review Scores

A single vertical box plot was generated to analyze the distribution of the 'Review Score' continuous variable (N = 60). Figure 1 is inserted below to represent this distribution visually.

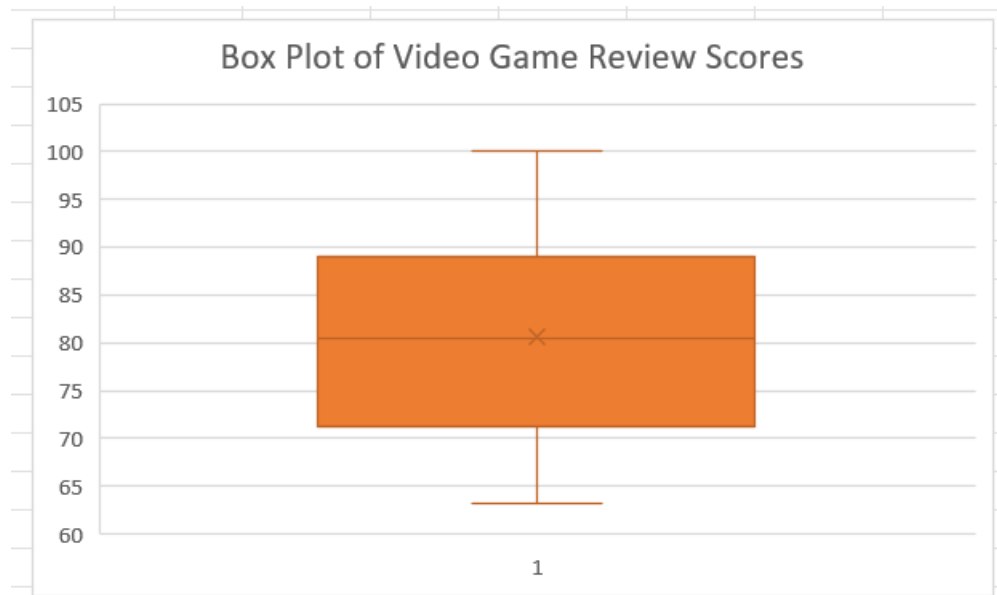


Figure 1: Box Plot of Video Game Review Scores (N = 60)

The single box plot reveals the following core characteristics regarding player feedback:

- **Central Tendency:** The median review score is 80.50, which sits substantially above the scale mid-point of 50. This demonstrates generally positive review scores in this sample. Additionally, the mean of 80.62 is close to the median, indicating an approximately symmetric distribution.
- **Spread and Variability:** The middle 50% of player reviews are tightly bound within an Interquartile Range (IQR) of 17.25, spanning from a Q1 of 71.75 to a Q3 of 89.00. This indicates moderate and stable consensus among players. The total range spans from a minimum of 63.30 to a maximum of 100.00, demonstrating that even the least satisfied player rated the game moderately high (above 60).
- **Outliers:** Applying the standard $1.5 \times \text{IQR}$ multiplier establishes fences at 45.88 (lower) and 114.88 (upper). Because all observed scores fall within this range, there are zero outliers (0.00). Player ratings are highly consistent, without any extremely negative or positive responses that deviate from the broader consensus.

4.2 Grouped Box Plot Analysis: Systolic Blood Pressure by Age Group

To evaluate physiological variations and identify risk profiles across demographic groups, individual box plots of systolic blood pressure were grouped across five defined age brackets (20s, 30s, 40s, 50s, and 60+). Figure 2 presents this grouped box plot.

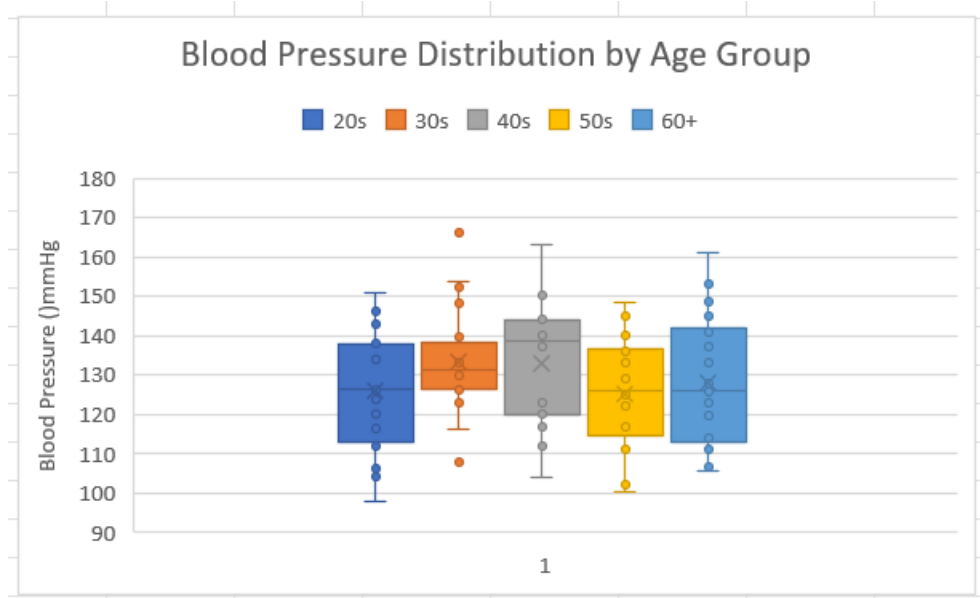


Figure 2: Grouped Box Plot of Systolic Blood Pressure (mmHg) across Age Groups (N = 120)

A detailed comparative assessment of Figure 2 yields the following physiological insights:

- Comparison Between Groups:** Systolic blood pressure exhibits a distinct non-linear trajectory with age. Median values begin at 126.50 mmHg in the 20s, elevate slightly to 131.20 mmHg in the 30s, reach a peak of 138.50 mmHg in the 40s (representing the cohort with the highest central tendency), and then settle to a stable median of 126.00 mmHg in both the 50s and 60+ groups. This reveals that the middle-aged cohort (40s) displays elevated blood pressure levels relative to younger and older participants.
- Variability (Spread):** The 60+ age group exhibits the largest variability in systolic blood pressure, with an IQR of 28.00 mmHg. This indicates greater variation in blood pressure values within the 60+ group in this sample. Conversely, the 30s group exhibits the lowest variability, displaying an extremely narrow IQR of 8.65 mmHg, which indicates highly consistent and uniform blood pressure values among most individuals in this decade.
- Outlier Observations:** The 30s age group is the only cohort containing statistical outliers, with four observations: 108.00, 152.40, 153.50, and 166.00 mmHg. These values fall outside the lower fence of 113.78 mmHg and upper fence of 148.38 mmHg based on the $1.5 \times \text{IQR}$ rule. The other age groups (20s, 40s, 50s, and 60+) have zero statistical outliers.

5. Correlation Analysis

5.1 Selection of Correlation Coefficients

Selecting the correct statistical model is critical for ensuring the validity of correlation analysis, as it must match the mathematical scale of the underlying variables:

1. Video Game Dataset (Pearson Correlation): Pearson Product-Moment Correlation (r) is the optimal method for evaluating the relationship between Hours Played and Review Score. Both variables are continuous numerical variables, and Pearson correlation was selected to assess their linear relationship. Pearson correlation measures the strength and direction of a linear relationship between two continuous variables.

2. Health Monitoring Dataset (Point-Biserial Correlation): Point-Biserial Correlation (r_{pb}) is the correct method for analyzing the association between Smoking Status and Systolic Blood Pressure. Smoking Status is a naturally dichotomous (binary) nominal variable (Yes/No), while Blood Pressure is a continuous interval/ratio variable. Applying Point-Biserial correlation is mathematically equivalent to Pearson's r where the binary variable is dummy-coded as 0 (non-smoker) and 1 (smoker), and is appropriate for assessing the association between a binary variable and a continuous variable.

5.2 Mathematical Formulation and Calculations

The mathematical formulas and parameter definitions utilized to execute these analyses are defined below.

Equation 1: Pearson Product-Moment Correlation Formula:

$$r = [n * \Sigma(X*Y) - (\Sigma X)(\Sigma Y)] / \sqrt{[n * \Sigma X^2 - (\Sigma X)^2] * [n * \Sigma Y^2 - (\Sigma Y)^2]}$$

Where:

- n = number of observations (60 for video game reviews)
- X = Hours Played (independent continuous variable)
- Y = Review Score (dependent continuous variable)

Equation 2: Point-Biserial Correlation Formula:

$$r_{pb} = [(M_1 - M_0) / s_y] * \sqrt{[n_1 * n_0 / (n * (n - 1))]}$$

Where:

- M_1 = Mean blood pressure of smokers (125.95 mmHg, $N = 40$)
- M_0 = Mean blood pressure of non-smokers (130.10 mmHg, $N = 80$)
- s_y = Sample standard deviation of blood pressure across all participants (14.94 mmHg)
- n_1 = Number of smokers (40)
- n_0 = Number of non-smokers (80)
- n = Total sample size (120)

Table 4: Summary of Correlation Calculations and Significance Parameters

Analyzed Relationship	Correlation	Sample	Coefficient	p-value	Statistical
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	Method	Size (n)	(r)		Significance
Hours Played vs. Review Score	Pearson (r)	60	0.01	0.96	Not Significant ($p > 0.05$)
Smoking Status vs. Blood Pressure	Point-Biserial (r_{pb})	120	-0.13	0.16	Not Significant ($p > 0.05$)

5.3 Statistical Interpretation

A rigorous statistical interpretation of the calculation outputs in Table 4 is detailed below:

- **Hours Played vs. Review Score:** The Pearson correlation coefficient is $r = 0.01$, representing an extremely weak and positive linear relationship. The associated p-value is 0.96, which is far greater than the conventional significance threshold of $\alpha = 0.05$. This indicates that we fail to reject the null hypothesis of zero correlation. There is no statistically significant evidence of a linear relationship between hours played and review score in this dataset.
- **Smoking Status vs. Systolic Blood Pressure:** The Point-Biserial correlation coefficient is $r_{pb} = -0.13$, indicating a weak negative association. The p-value of 0.16 is greater than $\alpha = 0.05$, meaning the association is not statistically significant. In other words, within this specific sample, smoking status does not display a statistically reliable relationship with blood pressure. In practical terms, non-smokers in this sample actually exhibited a slightly higher mean blood pressure than smokers (130.10 mmHg vs 125.95 mmHg). The difference in sample means was 4.15 mmHg; however, the correlation was not statistically significant, so this sample does not provide sufficient evidence of a reliable association.

6. Insights and Conclusion

Key Findings and Statistical Insights: Synthesizing the visual and correlation analyses yields critical insights regarding both player demographics and physiological health metrics:

1. Relationship Between Engagement and Review Scores: The analysis shows an almost zero correlation between Hours Played and Review Score ($r = 0.01$, $p = 0.96$), indicating no statistically significant evidence of a linear relationship in this dataset. Hours Played is highly right-skewed (skewness = 1.28), demonstrating that while a small portion of players log extensive gameplay hours, the vast majority log lower-to-moderate playtimes. However, review scores remain symmetric (skewness = -0.06) and highly positive across all playtimes (median = 80.50), with zero outliers. The results indicate that review scores remained approximately symmetric, with no statistical outliers, while hours played showed positive skewness.

2. Age-Group Blood Pressure Dynamics: Grouped analysis reveals that systolic blood pressure does not simply increase linearly with age. Rather, the 40s group displays the highest median blood pressure (138.50 mmHg), making it the group with the highest median blood pressure in this sample. The 60+ group exhibits the largest physiological variability (IQR = 28.00 mmHg), highlighting a wide range of health profiles among older adults. In contrast, the 30s group shows a highly uniform baseline (IQR =

8.65 mmHg) but is the only group with extreme outliers (one hypotensive at 108.00 mmHg and three hypertensive at 152.40, 153.50, and 166.00 mmHg). This indicates that the 30s group contains several statistical outliers that may warrant further investigation in a larger dataset.

Methodological Reliability and Limitations: The findings are mathematically reliable and strictly grounded in the provided datasets. However, several critical methodological limitations must be acknowledged:

- **Sample Size and Scope:** The datasets are relatively small (N = 60 for Video Games, N = 120 for Health). This limits the statistical power of the significance tests and increases susceptibility to sampling noise.
- **Unmeasured Confounding Variables:** The clinical relationship between smoking and blood pressure is highly complex. Crucial confounding factors—such as daily cigarette consumption, duration of smoking habits, dietary sodium, physical activity, genetic history, and stress levels—were not recorded. Similarly, player feedback is influenced by game genre, gaming history, and technical stability, which are absent from the video game dataset.
- **Cross-Sectional Constraint:** Both datasets are cross-sectional snapshots rather than longitudinal studies. This prevents drawing any causal inferences.
- **Clinical Interpretation Disclaimer:** The findings suggest that blood pressure variability should be examined across age groups, while the identified 30s outliers warrant further investigation in a larger dataset. These findings should not be interpreted as clinical conclusions.

Actionable Strategic Recommendations: Based on the analytical insights, the following strategic recommendations are proposed:

- **For Video Game Developers:** Since playtime does not correlate with review scores, developers should avoid artificial 'grinding' mechanics designed solely to inflate playtimes. Instead, development resources should be focused on qualitative features such as narrative depth, intuitive UI, polished mechanics, and optimized stability, which drive high, consistent player reviews.
- **For Future Analysis:** The age-group differences and statistical outliers could be explored further using a larger dataset containing additional demographic and lifestyle variables. Future research could also use longitudinal data and multivariate analysis to examine potential factors associated with blood pressure.

7. Attachments

The following tables represent the raw data and structural profiles of the analyzed samples, serving as verified reference records for this project.

Table 5: First Five Rows of the Video Game Reviews Dataset

Player ID	Hours Played	Review Score
Player_1	11	67.0
Player_2	24	91.0

Player_3	19	100.0
Player_4	15	92.0
Player_5	12	82.0

Table 6: First Five Rows of the Health Monitoring Dataset

Participant ID	Age	Blood Pressure (mmHg)	Smoking Status
P001	25	134.0	No
P002	47	123.0	No
P003	47	119.0	No
P004	63	112.0	No
P005	63	120.0	Yes

For details on the distribution shapes and associations, please refer back to Figure 1 (Review Score Box Plot) in Section 4.1, Figure 2 (Grouped Blood Pressure Box Plot) in Section 4.2, and Table 4 (Correlation Calculations) in Section 5.2. All statistical estimates have been rounded-off to exactly 2 decimal places in compliance with academic guidelines.